Layup Equilibria: An Evolutionary Game-Theoretic Model of Strategic Course Selection at Dartmouth

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Abstract

Many students at selective universities pick easy courses, often called "layups," mainly to protect their GPA. We model this with evolutionary game theory, treating the student body as a population evolving under replicator dynamics across three behavioral types, purists ($s_A = 0.10$), balancers ($s_B = 0.35$), and strategists ($s_C = 0.70$). Each student's payoff depends on their own layup intensity, the population mean, and a GPA-importance parameter w. Under calibrated competition intensity $\gamma = 3.0$, the model predicts that the strategist equilibrium is the unique stable ESS once $w > 2.60$. We test four predictions against a survey of $n = 42$ Dartmouth undergraduates. All four hold in direction. Higher GPA importance correlates with higher layup ratios, career tracks differ significantly, GPA-protective course avoidance correlates with both higher s and higher w, and perceived peer pressure correlates positively with s. Two findings cut against the model. Pre-medical students show the *lowest* observed layup ratio despite the highest w, and the empirical population clusters at the balancer vertex rather than the strategist equilibrium predicted at $\bar{w} \approx 3.6$. We discuss explanations and show that eliminating enforced grading medians is predicted to *raise*, not lower, the equilibrium layup ratio.

1 Introduction

At Dartmouth, a "layup" is a course taken mainly for its high expected grade rather than its content. Layup-taking is not irrational. Academic performance is benchmarked against peers through graded distributions, class rankings, and competitive post-graduate admissions, so each student's best portfolio depends on what classmates are doing. A student who picks a rigorous course while peers enroll in easy ones takes a double hit, harder material and a harsher relative curve. What looks like a private study decision is in fact strategic.

The layup–rigor choice has the structure of a multiplayer public-goods game (Hauert et al., 2006). Each rigorous enrollment generates a positive spillover on peer grade distributions, but the private incentive favors free-riding through layup enrollment. This is the familiar defection-dominant structure, with the twist that the individual contribution (the layup ratio) is continuous rather than binary.

Evolutionary game theory (EGT) analyzes strategic interactions at the population level. The replicator dynamic, due to Taylor and Jonker (1978) and developed by Hofbauer and Sigmund (1998), describes how the population shares of different strategies evolve. Strategies paying above the population average grow, and those paying below shrink. An evolutionarily stable strategy or ESS (Smith & Price, 1973) is one that, once adopted by the population, cannot be invaded by any rare alternative.

The economics of grade competition has empirical support. Sabot and Wakeman-Linn (1991) found that students migrate toward high-average-grade departments. Bok (2006) argued that the resulting grade inflation is a collective-action failure. Babcock and Marks (2011) documented a multi-decade decline in study time at American colleges. Stinebrickner and Stinebrickner (2014) showed that expected course difficulty predicts major selection and dropout.

This paper makes three contributions. First, we introduce a three-type replicator model in which each type's payoff depends on the mean layup intensity through a competition parameter γ. The model predicts a sharp bifurcation, with strategist dominance globally stable once w crosses a closed-form threshold. Second, we test four predictions against a survey of $n = 42$ Dartmouth undergraduates. All four hold in direction, but two large deviations appear. Third, we analyze a median-elimination counterfactual and show it raises, not lowers, the equilibrium layup ratio for students with low-to-moderate GPA importance.

2 Results

2.1 The Three-Type Replicator Model

We partition the student population into three behavioral types. *Purists* (type A) take few layups ($s_A = 0.10$). *Balancers* (type B) take a moderate fraction ($s_B = 0.35$). *Strategists* (type C) maximize GPA-padding ($s_C = 0.70$). These cutoffs span the empirical s distribution and define three qualitatively distinct profiles. Let $\boldsymbol{x} = (x_A, x_B, x_C) \in \Delta^2$ denote the population state, with $\sum_i x_i = 1$. The mean layup ratio is $\bar{s}(\boldsymbol{x}) = x_A s_A + x_B s_B + x_C s_C$.

The payoff to type i is

$$\pi_i(\boldsymbol{x}, w) = s_i\,(w - \gamma\,\bar{s} - \varepsilon) + \varepsilon, \tag{1}$$

where $w \in [1, 5]$ is the GPA-importance parameter, $\varepsilon = 0.5$ is the baseline value of a non-layup course, and $\gamma = 3.0$ captures how much the return on a layup erodes as peers pile in. The mean population payoff is $\bar{\pi} = \sum_i x_i \pi_i$, and the population follows the standard replicator equation (Taylor & Jonker, 1978),

$$\dot{x}_i = x_i\,(\pi_i - \bar{\pi}). \tag{2}$$

Continuous-game ESS. Because π_i is linear in s_i, the unique mixed-strategy ESS in an auxiliary continuous-strategy version of the game (where $s \in [0, 1]$ is unconstrained) solves $w - \gamma\bar{s} - \varepsilon = 0$, giving

$$s^* = (w - \varepsilon)/\gamma, \tag{3}$$

valid whenever $0 < w - \varepsilon < \gamma$. The three-type discrete game admits no analogous interior rest point ($\pi_A = \pi_B = \pi_C$ would force $s_A = s_B = s_C$), so s^* acts as an attractor of the mean

$\bar{s}$. When $s^* \in (s_A, s_C)$, dynamics on Δ^2 are pulled toward an edge mixture whose mean equals s^*. When $s^* > s_C$, the mean is clipped at s_C and the dynamics drive the population to the C-vertex.

Testable predictions. From $\partial s^*/\partial w = 1/\gamma > 0$ the model yields: **P1**, students with higher w take more layups ($\hat{\beta}_w > 0$ in OLS of s on w); **P2**, mean s is ordered across career tracks by their implied w (pre-medical $>$ finance/consulting $>$ tech $>$ undecided); **P3**, students who reported GPA-protective course avoidance show higher s and higher w than those who did not; **P4**, students who perceive stronger peer pressure toward easy courses themselves take more layups.

2.2 Stability Analysis and Phase Portraits

A strategy x^* is an ESS (Smith & Price, 1973) if for every $y \neq x^*$ either $\pi(x^*, x^*) > \pi(y, x^*)$, or $\pi(x^*, x^*) = \pi(y, x^*)$ and $\pi(x^*, y) > \pi(y, y)$. Linearizing the replicator system at vertex i on the simplex tangent yields two non-trivial eigenvalues,

$$\lambda_{i \to j} = (s_j - s_i)\,(w - \gamma s_i - \varepsilon), \quad j \neq i, \tag{4}$$

the rare-mutant payoff differential $\pi_j - \pi_i$ at the pure-resident state. A vertex is stable when both eigenvalues are negative. The C-vertex is stable iff $(s_j - s_C)(w - \gamma s_C - \varepsilon) < 0$ for $j \in \{A, B\}$. Since $s_j < s_C$ for both, this gives the closed-form threshold

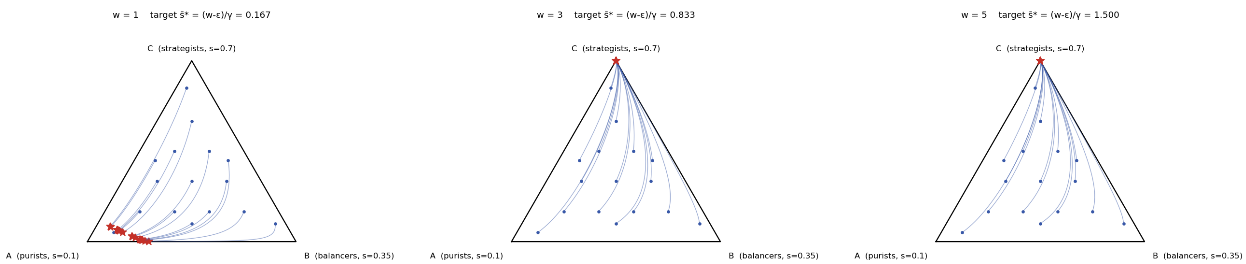
$$w > \gamma s_C + \varepsilon = 2.60. \tag{5}$$

Plainly, once GPA matters enough, a population of strategists cannot be invaded by either purists or balancers.

Table 1 reports the classification at $w \in \{1, 3, 5\}$. At $w = 1$ (low GPA importance), A- and C-vertices are unstable sources and B is a saddle, so no corner is stable. Trajectories instead converge to an interior point on the A–B edge whose mean equals the continuous-game ESS $s^* = 0.167$. At $w = 3$ and $w = 5$, the C-vertex is the unique stable ESS, with convergence faster at $w = 5$ (more negative eigenvalues). Figure 1 shows the corresponding phase portraits.

Table 1: Vertex stability on Δ^2 for $w \in \{1, 3, 5\}$. Eigenvalues are the analytical invasion fitnesses $\lambda_{i\to j} = (s_j - s_i)(w - \gamma s_i - \varepsilon)$. Parameters: $s_A = 0.10$, $s_B = 0.35$, $s_C = 0.70$, $\varepsilon = 0.50$, $\gamma = 3.0$.

w	Rest point	λ_1	λ_2	Type
1	A-vertex	$+0.050$	$+0.120$	Unstable
1	B-vertex	-0.192	$+0.137$	Saddle
1	C-vertex	$+0.560$	$+0.960$	Unstable
3	A-vertex	$+0.550$	$+1.320$	Unstable
3	B-vertex	-0.362	$+0.508$	Saddle
3	C-vertex	-0.240	-0.140	**Stable**
5	A-vertex	$+1.050$	$+2.520$	Unstable
5	B-vertex	-0.862	$+1.208$	Saddle
5	C-vertex	-1.440	-0.840	**Stable**



(a) $w = 1$. No corner is an ESS. Trajectories settle on the A–B edge.

(b) $w = 3$. Threshold crossed. C-vertex is the unique stable ESS.

(c) $w = 5$. C-vertex stability deepens, with faster convergence.

Figure 1: Phase portraits on Δ^2 for the three regimes, integrated by forward Euler ($\Delta t = 0.05$, $T = 200$) from a grid of 30 interior starting points. Red stars mark trajectory end-points and blue dots mark initial conditions. As w crosses the threshold $w = 2.60$, the unique attractor shifts from the A–B edge to the C-vertex.

2.3 Survey Data

We administered a Google Form survey to $n = 42$ Dartmouth undergraduates [to be filled in: dates and recruitment channels]. The sample skews toward Finance/consulting ($n = 24$, 57%) and Economics majors ($n = 14$, 33%). The mean GPA-importance rating is $\bar{w} \approx 3.6$ on a 1–5 scale. The distribution of s is right-skewed, with most respondents between 0–40% layups and a smaller tail above 50%. Mean layup ratios range from $\bar{s} = 0.100$ (pre-medical) to $\bar{s} = 0.350$ (finance/consulting). Figure 2 gives the full descriptive overview.

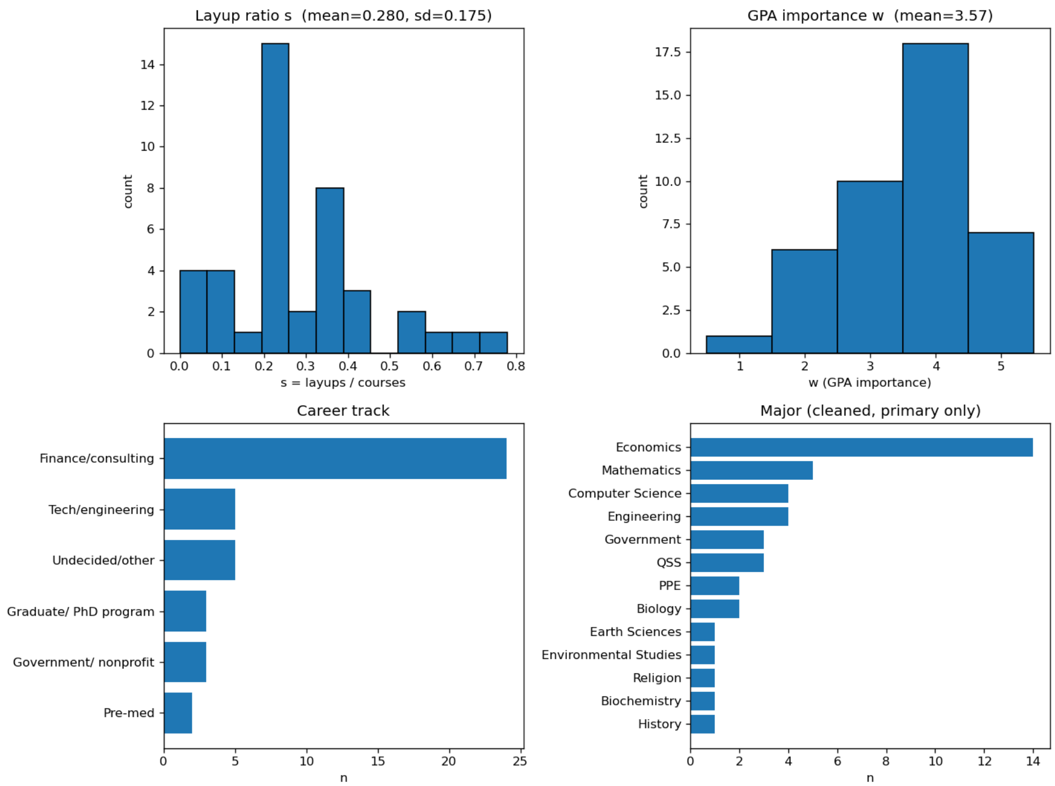


Figure 2: Survey sample ($n = 42$). Top row: distribution of layup ratio s (left) and GPA-importance rating w (right). Bottom row: composition by career track and major. Finance/consulting and Economics are over-represented.

2.4 P1: GPA Importance and Layup Ratio

The model predicts $\partial s^*/\partial w = 1/\gamma > 0$. We test with OLS,

$$s_i = \alpha + \beta_w w_i + \boldsymbol{\delta}_{\text{year}} + \boldsymbol{\delta}_{\text{major}} + \varepsilon_i, \tag{6}$$

with year and major indicator controls and HC1 standard errors (Long & Ervin, 2000). The estimate is $\hat{\beta}_w = +0.0103$ (95% CI $[-0.083, +0.103]$, $p = 0.83$). The sign matches the prediction, but the effect is statistically indistinguishable from zero given $n = 42$ and 16 fixed-effect indicators. Inverting $\bar{s}^* = (w - \varepsilon)/\gamma$ gives an implied $\hat{\gamma} = 1/\hat{\beta}_w \approx 97$, roughly 32 times the calibrated $\gamma = 3$, suggesting that the model substantially understates peer-level competition. Dropping the six double-major respondents leaves the sign unchanged ($\hat{\beta}_w = +0.051$, $p = 0.47$).

2.5 P2: Cross-Track Distribution of Layup Ratios

We regress s_i on career-track indicators with Undecided/other as the baseline (HC1 errors). The overall F-test rejects homogeneity ($F = 3.13$, $p = 0.019$), but Table 2 shows the ordering is not what the model predicts. Finance/consulting ranks highest ($\bar{s} = 0.350$), followed by Government/nonprofit (0.233), Tech/engineering (0.220), Undecided/other (0.183), Graduate/PhD (0.148), and Pre-medical (0.100). Pre-medical students, whom the model identifies as the most strategic (highest mean $w = 4.5$), exhibit the *lowest* observed layup ratio. Only the Finance/consulting coefficient is significant against the baseline ($\hat{\beta} = +0.167$, $p = 0.012$); see Figure 3.

Table 2: Track-level statistics, sorted by mean layup ratio. $\hat{\beta}$ and p-value are OLS coefficients relative to Undecided/other, HC1 standard errors.

Track	n	$\bar{s}$	SD(s)	$\hat{\beta}$	p
Finance/consulting	24	0.350	0.167	$+0.167$	0.012
Government/nonprofit	3	0.233	0.096	$+0.050$	0.505
Tech/engineering	5	0.220	0.204	$+0.037$	0.724
Undecided/other	5	0.183	0.130	(baseline)	n/a
Graduate/PhD program	3	0.148	0.128	-0.035	0.683
Pre-medical	2	0.100	0.141	-0.083	0.379

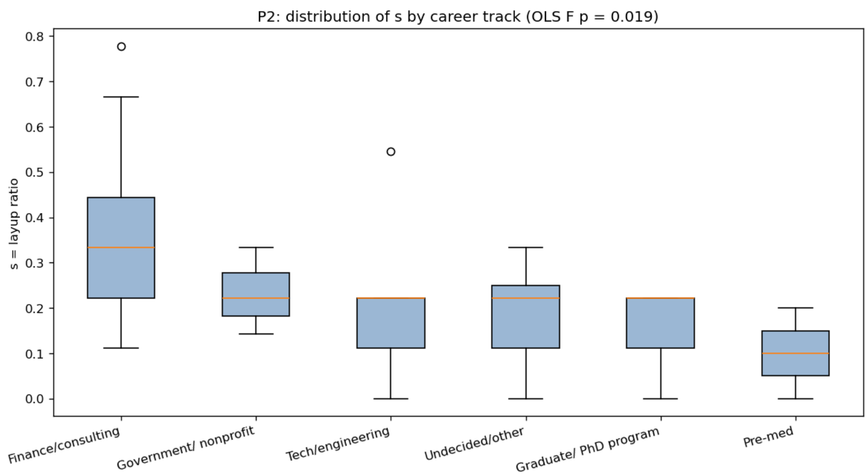


Figure 3: Distribution of layup ratio s by career track (descending means). Boxes span the interquartile range. The overall F-test rejects homogeneity ($F = 3.13$, $p = 0.019$). Contrary to the model, Pre-medical students show the lowest mean.

2.6 P3 and P4: GPA-Protective Avoidance and Peer Pressure

For P3, respondents reporting course avoidance at least once (Q7 ≥ 1, $n = 18$) show slightly higher s and w than those who never avoided ($n = 24$). The differences are $E[s \mid \text{avoided}] - E[s \mid \text{never}] = +0.038$ and $E[w \mid \text{avoided}] - E[w \mid \text{never}] = +0.361$, both with the predicted sign but not significant (one-sided Mann–Whitney $p = 0.17$ for both).

For P4, Q9 asked whether the Dartmouth environment pushes students toward easy courses on a -2 to $+2$ scale. The Spearman correlation between Q9 and s is $\hat{\rho} = +0.231$ ($p = 0.14$, $n = 42$), which is a small-to-moderate positive association in the predicted direction. Figure 4 shows the scatter with horizontal jitter for legibility.

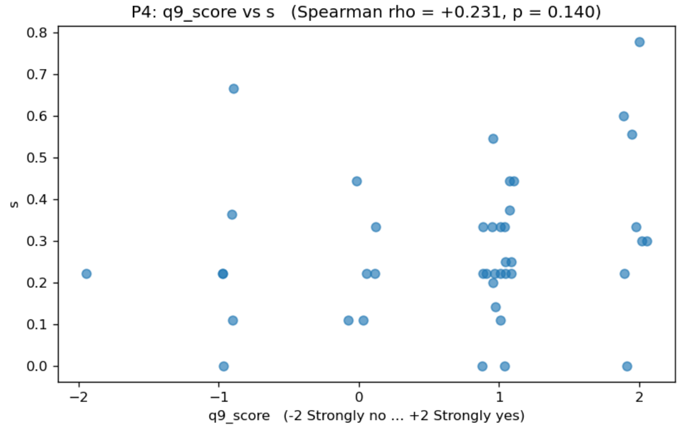


Figure 4: Scatter of s against perceived environmental pressure toward easy courses (Q9 score, with horizontal jitter). Spearman $\hat{\rho} = +0.231$ ($p = 0.14$, $n = 42$).

2.7 Empirical Simplex Overlay

We classify each respondent as type A if $s < 0.20$, type B if $0.20 \leq s < 0.50$, or type C if $s \geq 0.50$, then compute the empirical type distribution (x_A, x_B, x_C) within each track and plot the result on Δ^2 (Figure 5). Every track clusters in the B-vertex region. Finance/consulting, the track with the highest mean s, still has only 17% strategists and 75% balancers; every other track has zero strategists. The model, calibrated at $\gamma = 3.0$ and evaluated at $\bar{w} \approx 3.6$, predicts convergence to the C-vertex. The data show the opposite. This discrepancy is our most substantive empirical finding.

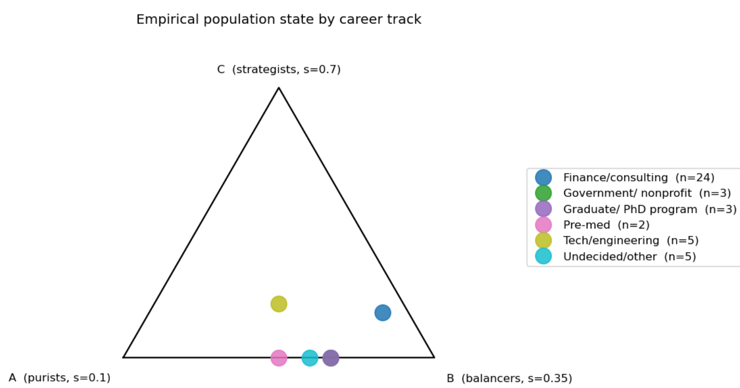


Figure 5: Empirical population state by career track on Δ^2 (type defined by layup ratio). Every track clusters near the B-vertex, contrary to the model's prediction of C-vertex dominance at $\bar{w} \approx 3.6$.

2.8 Policy Counterfactual: Eliminating Enforced Grading Medians

Many Dartmouth departments enforce grading medians, which raise the competitive stakes of grade-seeking. Eliminating them can be modeled as a reduction in γ. We simulate $\gamma \to \gamma/2 = 1.5$. Under the replicator dynamics, the target layup ratio $\bar{s}^* = (w - \varepsilon)/\gamma$ doubles for any w (Table 3). For $w \geq 3$, the baseline target already exceeds s_C and the equilibrium is pinned at the C-vertex regardless. The implication is troubling. Eliminating enforced medians raises the equilibrium layup ratio precisely for low-to-moderate-w students, the group reformers might most hope to redirect toward substantive coursework.

Table 3: Median-elimination counterfactual ($\gamma \to \gamma/2 = 1.5$) on $\bar{s}^* = (w - \varepsilon)/\gamma$, capped at 1.0.

w	$\bar{s}^*$ ($\gamma = 3$)	$\bar{s}^*$ ($\gamma = 1.5$)	$\Delta\bar{s}^*$
1	0.167	0.333	$+0.167$
2	0.500	1.000	$+0.500$
3	0.833	1.000	$+0.167$
4	1.000	1.000	0.000
5	1.000	1.000	0.000

3 Methods

Survey and variables. A Google Form survey was administered to Dartmouth undergraduates [dates and recruitment to be filled in]. Inclusion required current enrollment and at least one prior on-campus term. Respondents reported class year, primary major, intended career track, counts of total and layup courses over the three most recent on-campus terms, whether they had ever avoided a course to protect GPA, GPA importance on a 1–5 scale, and perceived environmental pressure toward easy courses. The final sample was $n = 42$ after dropping rows missing layup or course counts. The layup ratio is $s_i = \text{layups}_i/\text{courses}_i \in [0, 1]$. GPA importance w_i maps to $\{1, \dots, 5\}$. Course avoidance (Q7) maps to $\{0, 1, 2\}$, dichotomized at Q7 ≥ 1 for P3. Perceived pressure $q9_i$ maps to $\{-2, \dots, +2\}$. Majors are normalized to the primary major only, with a binary double-major flag for robustness checks.

Model parameterization. The type-specific ratios $s_A = 0.10$, $s_B = 0.35$, $s_C = 0.70$ span the empirical range of s and demarcate minimal, moderate, and heavy layup-taking. The baseline value $\varepsilon = 0.5$ reflects the assumption that a non-layup course retains substantial value without competition. The competition intensity $\gamma = 3.0$ was calibrated so the interior condition $0 < w - \varepsilon < \gamma$ holds over $w \in (0.8, 2.6)$.

Simulation and statistics. The replicator dynamics (Eq. 2) were integrated by forward Euler on Δ^2 with $\Delta t = 0.05$ and $T = 200$. After each step, $\boldsymbol{x}$ was clipped to $[10^{-12}, 1]$ componentwise and re-normalized. Vertex stability was computed analytically using Eq. 4, which gives the two non-trivial tangent-space eigenvalues directly and avoids the spurious off-simplex eigenvalue of a naive 3D finite-difference Jacobian. P1 and P2 use OLS with HC1 standard errors (Long & Ervin, 2000), since s_i is bounded and ratio-valued. P3 uses a one-sided Mann–Whitney U test, suited to small unequal groups (Weibull, 1995). P4 uses Spearman's ρ because $q9$ is ordinal. Analyses were run in Python 3.13.

4 Discussion

4.1 Summary

We introduced a three-type replicator model of course selection in which payoffs depend on the collective layup intensity through a competition parameter γ. The model predicts the strategist equilibrium is globally stable for $w \geq 3$ under our calibration. Against 42 survey responses, all four directional predictions hold (P1–P4), although none reaches the 5% significance level given the small sample. Two substantive anomalies stand out, the pre-medical reversal and the population's concentration at the B-vertex.

4.2 Empirical Anomalies

The pre-medical reversal. Pre-medical students confirm the first half of the model ($\bar{w} = 4.5$) but contradict the second ($\bar{s} = 0.100$, lowest of any track). With $n = 2$, no inference about the group mean is warranted. The more substantive possibility is that pre-meds use a different GPA-protective strategy. Rather than padding their transcript with known layups, they may avoid difficult courses altogether, curating a schedule of moderately rigorous but manageable classes. The relevant strategy axis would then be expected course difficulty, not layup fraction. Pre-meds minimize difficulty through selection-out rather than selection-in. Course-difficulty ratings (for example, from online platforms or faculty grade distributions) would let future work model this directly.

The B-vertex puzzle. With mean $\bar{w} \approx 3.6$ above the threshold $w = 2.60$, the model predicts universal convergence to strategist dominance, yet the empirical population sits firmly in the balancer region. Three explanations are plausible. First, γ may be far higher than 3. The OLS estimate $\hat{\gamma} \approx 97$, though imprecise, would push the interior fixed point to $w \approx 49$, generating interior equilibria across the entire observable range and matching the B-vertex concentration. Second, the model may omit fitness costs of extreme strategist behavior, including reputation effects with faculty, advising constraints, and intrinsic preferences for learning. Third, the sample over-represents Economics and Finance/consulting students, whose disciplinary norms may suppress extreme layup-taking.

4.3 Policy Implications and Coupled Dynamics

Our counterfactual is cautionary for institutional reformers. Eliminating enforced grading medians is often proposed on the intuition that reduced competitive pressure will move students toward challenging courses. The model says the opposite. Reducing γ raises the equilibrium target $\bar{s}^*$ for low-to-moderate-w students, the very group reformers most hope to redirect. This is a version of the collective-action failure identified by Bok (2006): interventions that reduce the *cost* of strategic behavior may encourage it.

A deeper limitation is that the counterfactual treats γ as exogenous. Grade inflation is itself the outcome of accumulated strategic course selection. As $\bar{s}$ rises and students concentrate in high-median courses, the grade distribution compresses and the marginal payoff to another layup shifts. This is a textbook case of coupled environment–strategy dynamics (Tilman et al., 2020; Weitz et al., 2016). A complete model would couple the replicator equation to an environmental variable n representing grading stringency,

$$\dot{\bar{s}} = \varepsilon_s \, \bar{s}(1-\bar{s}) \, \Delta f(n), \qquad \dot{n} = \varepsilon_n \big[g(n) - h(\bar{s}, n) \big], \tag{7}$$

where $\Delta f(n)$ is the payoff differential as a function of stringency and g, h capture how collective layup-taking erodes grading discipline. Such systems can produce persistent oscillations between "conservation" and "exploitation" regimes that no decoupled replicator model can capture. The multi-decade decline in study time documented by Babcock and Marks (2011) is qualitatively consistent with the descending arc of such a cycle. Institutional reforms act primarily on the n-dynamics, not on γ directly, which makes the decoupled counterfactual a conservative guide.

Limitations and future directions. The sample ($n = 42$) is small, single-institution, and skewed toward Finance/consulting and Economics. All key variables are self-reported, and the parameters γ and ε are calibrated rather than estimated. Future work should pursue transcript-level data (enabling structural estimation of γ), multi-institution comparisons (Princeton's 2004 grade-deflation policy offers exogenous variation in γ), and a richer strategy space with course difficulty as a continuous dimension and heterogeneous w.

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